

M.D.S.SUBRAHMANYAM

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EDUCATION

- Kalsalingam Academy Of Research And Education** TamilNadu, India
Computer Science and Engineering(AIML); GPA: 8.48 *July 2022 - April 2026*
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases

SKILLS SUMMARY

- Languages** Python, C++, JavaScript, SQL
- Frameworks** TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy.
- Tools** Git, GitHub, Jira, Docker
- Web Development** HTML, CSS, JavaScript, Flutter, Flask.
- Platforms** Linux, Arduino, Docker
- Soft Skills** Communication, Teamwork, Problem-Solving, Adaptability, Time Management, Quick Learning.

EXPERIENCE

- LG Soft India** Bengaluru
Research Engineer Intern *July 2025 – February 2026*
 - Contributed to the development and validation of embedded audio systems in LG Smart TVs as part of the Audio Renderer team in the BSP division.
 - Debugged and resolved 10+ audio pipeline defects across HDMI audio-video sync, volume control logic, and codec compatibility on LG Smart TV embedded systems, improving playback reliability in QA regression cycles.
 - Analyzed end-to-end audio data flow and signal processing pipelines in embedded multimedia firmware, identifying 5 synchronization bottlenecks and documenting root causes.
 - Contributed to validation of the Personalized Sound Wizard — an AI-driven adaptive audio optimization module — by designing and executing 30+ test cases covering edge conditions across 3 Smart TV SKUs.

PROJECTS

- Web Platform for Recipe Scaling and Per-Serving Ingredient Delivery with AI-Powered Cooking Assistance:** Developed a full-stack web platform enabling **per-serving ingredient delivery**, **dynamic recipe scaling**, and **real-time customization**, integrated with an **NLP-based AI cooking assistant**. The system bridges **recipe planning and ingredient delivery** by providing **precise, personalized quantities**, reducing food waste. Designed a **modular (MCP-style) backend architecture** to coordinate **recipe management, ingredient scaling, order processing, and AI chatbot services** via REST APIs for **scalable and efficient communication**. Published at **IEEE ICoECIT 2026**.
Tech: React, Node.js, Express, PostgreSQL, NLP-based AI Chatbot, [GitHub](#).
- Brain Tumor Detection:** Developed a binary MRI classifier (tumor / non-tumor) using **VGG19** fine-tuned via transfer learning on ImageNet weights. Applied data augmentation and class-balanced sampling to address dataset imbalance, achieving **83% accuracy** and **87% sensitivity** on tumor-positive cases. Outperforming a baseline CNN trained from scratch by 9 percentage points. Designed to support early-stage medical diagnosis workflows.
Tech: Python, Pandas, Tensor-Flow, Open CV, [GitHub](#).
- Real-Time Vehicle Detection & Speed Estimation System:** Developed a real-time vehicle detection and tracking system using **YOLOv11** and **Enhanced SORT**, achieving **~30 FPS** on 720p video streams. Implemented motion prediction with Kalman Filter, IOU matching, and the **Hungarian Algorithm** for stable multi-object tracking. Developed camera calibration and speed estimation using pixel-to-meter conversion and statistical filtering, generating structured traffic analytics (CSV/JSON). **Published at IEEE ICAFT 2025**.
Tech: Python, HTML, CSS, JavaScript, YOLOv11, OpenCV, [GitHub](#).
- Smart Classroom Lighting System:** Developed an occupancy-based classroom light automation system using **2 PIR sensors**, **Arduino Uno**, and a **5V relay module**. Detects entry/exit direction to maintain real-time occupant count, enabling automatic ON/OFF lighting control with **~30–40% reduction** in unnecessary power consumption.
Tech: Python, Arduino IDE, C++, IOT- Hardware, [GitHub](#).

HONORS AND AWARDS

- Achieved 2nd Prize at ApexCode for delivering an innovative and high-performing solution.
- Awarded 4th Prize for presenting an innovative IoT concept.
- Achieved a Top 5 finish in the IBM Hackathon for delivering an impactful project solution.
- Completed an online internship at Rinex On Artificial Intelligence , where I worked on foundational concepts and practical applications in Artificial Intelligence and Machine Learning.

PUBLICATIONS

- IEEE ICAFT 2025** — Vehicle Detection and Speed Estimation using YOLO v11 (IEEE Xplore).
- Published an **IEEE Xplore-indexed research paper** titled “A Web Platform for Recipe Scaling and Per-Serving Ingredient Delivery with AI-Powered Cooking Assistance” at the **IEEE International Conference on Emerging Computing and Intelligent Technologies (ICoECIT 2026)**.